```
> restart;#The momopole RadioFrequency Planck's "constant"
> Digits:=25;
```

$$Digits := 25 \tag{1}$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, Tq\, A\, \nu + 1) + C\, \nu^2 \tag{2}$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=3.693*10^(-18);
  ec:= 1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 3.693000000000000000000000 \times 10^{-18}$$

$$ec := 1.602176634000000000000000 \times 10^{-19} \tag{3}$$

```
> A0:=64/3*Tq;
```

$$A0 := 3.765333333333333333333333 \times 10^{-18} \tag{4}$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 3.765333333333333333300000 \times 10^{-18} A\, \nu) + C\, \nu^2 \tag{5}$$

```
> E1:=subs(nu=5.754173859*10^8,E(nu));
```

$$E1 := Em + E0 \ln(1 + 2.166638263708799999980819 \times 10^{-9} A) + 3.311051679959895188100000 \times 10^{17} C \tag{6}$$

```
> E2:=subs(nu=6.338671958*10^8,E(nu));
```

$$E2 := Em + E0 \ln(1 + 2.386721281252266666645538 \times 10^{-9} A) + 4.017876219113555376400000 \times 10^{17} C \tag{7}$$

```
> E3:=subs(nu=31*10^9,E(nu));
```

$$E3 := Em + E0 \ln(1 + 1.167253333333333333323000 \times 10^{-7} A) + 961000000000000000000 C \tag{8}$$

```
> solve({E1=3.259,E2=3.59});
```

$$\{A = A, C = 1.414778271842943053715718 \times 10^{-17} E0 \ln(1. + 2.166638263708799999980819 \times 10^{-9} A) - 1.414778271842943053715718 \times 10^{-17} E0 \ln(1. + 2.386721281252266666645538 \times 10^{-9} A) + 4.682916079800141507799027 \times 10^{-18}, E0 = E0, Em = 4.684403973756333877677291\, E0 \ln(1. + 2.386721281252266666645538 \times 10^{-9} A) - 5.684403973756333877677291\, E0 \ln(1. + 2.166638263708799999980819 \times 10^{-9} A) + 1.708462284686653486488817\} \tag{9}$$

```
> sol:=solve({E1=3.259,E2=3.59,E3=0},{Em,E0,C,A});
```

$$sol := \{A = A, C = (0.10000000000000000000000000\ (3259. \ln(1. \tag{10}$$

$$+ 2.38672128125226666645538 \times 10^{-9} A) + 331. \ln(1. + 1.16725333333333333323000$$

$$\times 10^{-7} A) - 3590. \ln(1. + 2.16663826370879999980819 \times 10^{-9} A))) /$$

$$(7.068245391536601883 \times 10^{18} \ln(1. + 1.16725333333333333323000 \times 10^{-7} A)$$

$$- 9.606688948320040104811 9 \times 10^{22} \ln(1. + 2.38672128125226666645538 \times 10^{-9} A)$$

$$+ 9.605982123780886444623 6 \times 10^{22} \ln(1. + 2.16663826370879999980819 \times 10^{-9} A)),$$

$$E0 = -(3.18211758306703505324640 9 \times 10^{22}) / (7.068245391536601883 \times 10^{18} \ln(1.$$

$$+ 1.16725333333333333323000 \times 10^{-7} A) - 9.606688948320040104811 9 \times 10^{22} \ln(1.$$

$$+ 2.38672128125226666645538 \times 10^{-9} A) + 9.605982123780886444623 6 \times 10^{22} \ln(1.$$

$$+ 2.16663826370879999980819 \times 10^{-9} A)), Em$$

$$= (0.00200000000000000000000000 (6.037915335175266232043 \times 10^{21} \ln(1.$$
$$+ 1.16725333333333333323000 \times 10^{-7} A) - 1.56594950000000000000000 \times 10^{26} \ln(1.$$
$$+ 2.38672128125226666645538 \times 10^{-9} A) + 1.72499500000000000000000 \times 10^{26} \ln(1.$$
$$+ 2.16663826370879999980819 \times 10^{-9} A))) / (7.068245391536601883 \times 10^{18} \ln(1.$$
$$+ 1.16725333333333333323000 \times 10^{-7} A) - 9.606688948320040104811 9 \times 10^{22} \ln(1.$$
$$+ 2.38672128125226666645538 \times 10^{-9} A) + 9.605982123780886444623 6 \times 10^{22} \ln(1.$$
$$+ 2.16663826370879999980819 \times 10^{-9} A))\}$$

```
> A:=6993.8*10^4;evalf(sol[4]);#Em
```

$$A := 6.99380000 \times 10^{7}$$

$$Em = -0.28246484557430985473556 74 \quad (11)$$

```
> Em:=-0.28246484557430985473556 74;
```

$$Em := -0.28246484557430985473556 74 \quad (12)$$

```
> evalf(sol[3]);#E0=1.722290354;
```

(13)

$$E0 = 25.236257701193162341134 93 \quad (13)$$

```
> E0:=25.23625770119316234113493;
```

$$E0 := 25.236257701193162341134 93 \quad (14)$$

```
> evalf(sol[2]);#C
```

$$C = -5.787898549832585630549072 \times 10^{-20} \quad (15)$$

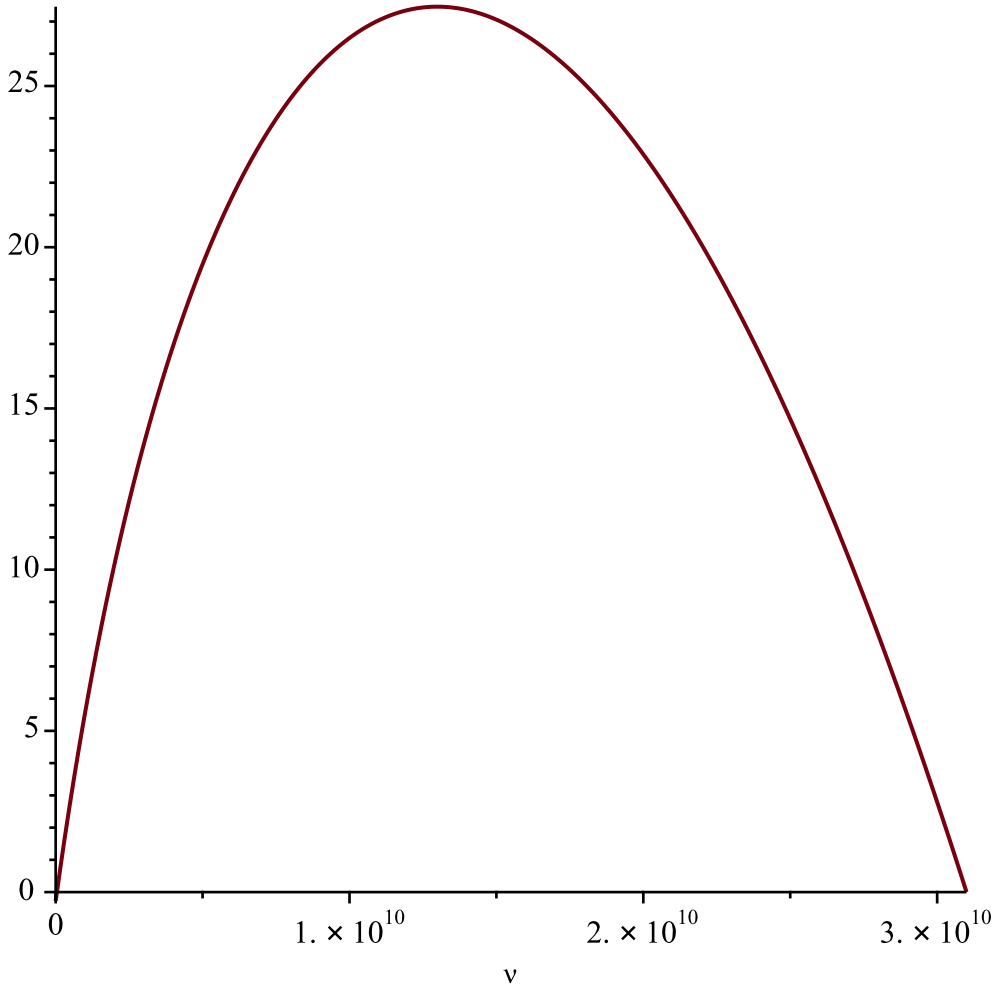
```
> C:=-5.787898549832585630549072*10^(-20);
```

$$C := -5.787898549832585630549072 \times 10^{-20} \quad (16)$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := -0.282464845574309854735 5674 + 25.236257701193162341134 93 \ln(1 + 2.633398826666666666643354 \times 10^{-10} \nu) - 5.787898549832585630549072 \times 10^{-20} \nu^2 \quad (17)$$

```
> plot(E(nu),nu=10^6..3.1*10^10);
```




```
> evalf(subs(nu=5.754173859*10^8,E(nu)));evalf(subs(nu=6.338671958*
  10^8,E(nu)));evalf(subs(nu=6.974105195*10^8,E(nu)));
```

3.258999999999999999999966

3.589999999999999999999962

(18)

$$3.944417754026645618964373 \quad (18)$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=42.7*10^6,E(nu)));evalf
  (subs(nu=15.4*10^9,E(nu)));#26.9eV @ 15.9Ghz in the range
  [42Mhz-31Ghz]
```

$$-0.28246484557430985473556 74$$

$$-0.00038201727125452853049 39$$

$$26.8853955564658593409780 2 \quad (19)$$

```
> mu:=4*Pi*10^(-7);rb:= 5.292410932*10^(-11);solve((Pi*mu*E0m*rb^3)
  ^0.5=2.360457908*10^(-18));
```

$$\mu := \frac{\pi}{2500000}$$

$$rb := 5.29241093200000000000000 \times 10^{-11}$$

$$9.52077080897457803652756 6 \quad (20)$$

```
> h := 6.62607015*10^(-34);ec:= 1.602176634*10^(-19);he:=h/ec;rm:=
  5.64*10^(-21);Bmw:=mu*B*h/(8*Pi*rm^2);mu*B*he/(8*Pi*rm^2);E0m:=
  25.23625770119316234113493;Bmw/1.748;(8*Pi/3*mu*E0m*rb^3)^0.5/
  (Pi*mu*E0m*rb^3)^0.5;
```

$$h := 6.62607015000000000000000 \times 10^{-34}$$

$$ec := 1.60217663400000000000000 \times 10^{-19}$$

$$he := 4.13566769692385864616223 1 \times 10^{-15}$$

$$rm := 5.64000000000000000000000 \times 10^{-21}$$

$$Bmw := 3.84633523589576731552738 9 \times 10^{-18}$$

$$24.00693627826160967172980$$

$$E0m := 25.23625770119316234113493$$

$$2.20042061550101105007287 7 \times 10^{-18}$$

$$1.63299316185545206546485 6 \quad (21)$$

```
> (Pi*mu*E0m*rb^3)^0.5;(8*Pi/3*mu*E0m*1.478*rb^3)^0.5;(8*Pi/3*mu*
  E0m*rb^3)^0.5/(Pi*mu*E0m*rb^3)^0.5;Bm:=Bmw/1.478;Bm/1.478;
```

$$3.84302032980382992140762 9 \times 10^{-18}$$

$$7.62946816273358593099409 5 \times 10^{-18}$$

$$1.63299316185545206546485 6$$

$$Bm := 2.60239190520687910387509 4 \times 10^{-18}$$

$$1.76075230392887625431332 5 \times 10^{-18} \quad (22)$$